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Productive Intelligence: AI and Generative AI for Production Optimization and Predictive Maintenance in Oil and Gas Fields – Case Studies from the USA and India

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Abstract

This paper describes real-world views of how Artificial Intelligence (AI) and Generative AI (Gen AI) can be employed to enhance oil and gas well production optimization and predictive maintenance. We employed machine learning algorithms to predict and analyze day-to-day production, pressure, and equipment sensor readings. Generative models were used in parallel to simulate possible failure modes. From India and US case studies, we demonstrate how smart systems perceive and facilitate decision-making in order to improve production efficiency. Implementation of the methodology provided measurable improvement in performance. Downtime resulting from failures due to maintenance was predicted 15 to 20 percent more accurately and there was reduction in unplanned downtime by up to 18 percent. Overall, the findings show predictive analytics with generative simulation is a battle-tested, scalable solution that maximizes reliability and optimizes production for many operating conditions. They operated by taking in historical and real-time well data into machine learning models that were programmed to pick up on patterns that might otherwise be overlooked through traditional analysis. Gen AI, when married with digital twin-based modeling, allowed for predictive maintenance and failure scenario simulation, not only making operations smarter but also more responsive. Secondly, through the use of edge-computing models, the information produced could be retrieved nearly instantaneously by field personnel, enhancing response time to anomalies and maximizing daily operational tactics. This paper provides suggestions as to how contemporary AI methodologies can be adapted to the actual issues encountered in upstream oilfield activities, particularly in operations with sensor reliability problems, excessive gas-to-oil ratios, pump inefficiency, and challenging production conditions. Through three different case studies and proof of quantifiable performance improvement, we lay a compelling basis for incorporating AI and Gen AI into current digital transformation initiatives in the petroleum industry.

Introduction

The oil and gas industry is facing a revolutionary shift, with rising complexity of field operations and mounting requirements on efficient resource management. The conventional and unconventional reservoirs offer to the operator's constant production inefficiencies, unpredictable equipment behavior, and growing

costs of maintenance. The traditional monitoring techniques rely on periodic human data collection and maintenance-oriented repair cycles, making it difficult to manage failure prediction or optimize production in real-time.

Over the past few years, the sector has been shifting away from reactive processes to predictive and prescriptive ones. Production activities are increasingly demanding scalable systems with the capability to identify anomalies, forecast downtime incidents, and issue recommendations in real-time. In older wells and remote production areas particularly, not having continuous monitoring and relying on delayed manual measurements lead to problems remaining undiscovered until they fail. This has a direct impact on operating expenses (OPEX), safety, and asset longevity. Artificial Intelligence (AI) application has gained momentum in midstream and upstream operations. AI enables efficient means of uncovering the hidden patterns of production data, machine behavior, and making decisions at a faster rate. With abundant data from SCADA systems, sensors, and logs, AI applications assist in giving priority to focus, finding windows of opportunity to optimize, and minimizing intervention time. Generative AI (Gen AI) has recently emerged as another technology with the capability to mimic operating conditions and real-time inputs.

In contrast to the conventional AI models dependent on classification or regression, Gen AI is capable of simulating operating scenarios, predicting sequences of failure, and creating actionable recommendations even in the case of data-poor scenarios. This renders it suitable for sophisticated upstream scenarios where every well acts differently. This article announces a real-world deployment of Gen AI and AI solutions to optimize production and perform predictive maintenance for oil and gas wells. Through the fusion of machine learning models, we reached a methodology that delivers near real time monitoring and automated intelligence. We tested the methodology based on field data for producing assets in India and the United States to demonstrate it under various geologic and operating conditions. The objective of this study is to show how AI-based techniques can optimize production performance, reduce unscheduled downtime, and facilitate improved day-to-day field decision-making. We present three onshore well case histories where using AI and Gen AI technology led to measurable improvement in forecast accuracy and operating results. With this research, we hope to present a field-tested and replicable model that enables digital transformation for oil and gas operations. Not only is it a technical development, but it is also an advancement towards more sustainable and proactive field management.

Methodology

The process described here integrates Artificial Intelligence (AI), Generative AI (Gen AI), and edge computing to assist in optimizing production and predictive maintenance of oil and gas fields. The platform was built to analyze multi-source production operation data from production wells, including time-series production rates, surface pressure readings, and sensor logs on equipment. The process integrates data-driven machine learning models with generative simulation tools to enable real-time monitoring, early anomaly detection, and scenario-based forecasting. The AI pipeline combined gradient boosting regression, random forests, and clustering algorithms to capture both predictive and anomaly detection tasks. Time-series models, including LSTM variants, were evaluated for recurring pressure fluctuations. Feature engineering focused on operationally relevant parameters such as tubing and casing pressures, fluid levels, pump speeds, and GOR values, ensuring only the most significant predictors were used. Supervised and unsupervised machine learning techniques were used to build AI models. Historical production data were used to train the models to learn patterns of performance, identify anomalies, and flag warning signs of equipment degradation or production inefficiency. Feature engineering techniques were utilized to create insightful signals from pressure measurements, production rates, and operating parameters. Anomaly detection software such as clustering and statistical control models were used to detect abnormal deviations in well performance or equipment operation. In this deployment, clustering models were utilized to segment

non-evident outliers in flow rates and pressure trends, while gradient-boosting models were used to facilitate improved performance prediction.

The models were validated with cross-validation and optimized by hyper parameter optimization routines to make them applicable for new wells and operating conditions. Parallely, Generative AI models were utilized to model failure propagation paths and generate prescriptive maintenance scenarios. Trained on the sensor patterns and field maintenance histories, the models were used to investigate "what-if" scenarios under different pressure, flow rate, or equipment stress conditions. This enabled field engineers to visualize probable locations of failure before they happen and plan remedial action in advance. A digital twin architecture was also used to model the operational performance of every well. These twins were being updated in near real time from data taken in the field and post-processed by edge-deployed inference engines. This provided on-site decision-making that is independent of central systems, minimizing latency and enabling quicker responsiveness to abnormal conditions.

To enhance reliability and minimize noise, the data pipelines further incorporated preprocessing operations like outlier filtering, imputation for missing readings, and time-window smoothing. Sensor readouts were analyzed for health and drift, and malfunctioning channels (e.g., stuck SICP values) were marked automatically. A dash-based cloud was built to bridge the distance between the AI insights and operators and engineers to see KPIs, get automated alerts, and communicate with generative simulations. Multi-well performance comparisons and field-wide optimization plans were made possible by the platform by bringing together the streams of real-time and history data. It gave rise to a robust platform for decision support and predictive analytics by putting together machine learning, Gen AI, digital twins, and edge-cloud infrastructure. It made way for a transition from reactive, manual operations to a more autonomous, proactive production system, which could be tunable across varied geologic and geographic terrains. To enhance adaptability, transfer learning approaches were applied so that models trained on data-rich wells could be fine-tuned for wells with limited data. Generative simulations also helped augment sparse datasets, ensuring reliable predictions even under low-data conditions. The system was built to scale across assets, independent of the maturity of the field or hardware infrastructure.

Field Case Studies

SW Louisiana Well (KH#1)

A production well in Southwest Louisiana was selected to demonstrate the initial application of AI and Gen AI for production diagnostics. This well was selected because it represented a moderately mature asset with a relatively stable oil output but recurring water production anomalies, making it an ideal candidate for early anomaly detection testing. The operator provided 55 days of daily production data, including oil and water volumes, tubing pressure, and surface casing pressure. A few days of data were not provided as the well was shut in for maintenance. Initial analysis found pressure readings to be generally constant, with the SICP (Shut-In Casing Pressure) sensor stuck at 125 psi on nearly all days, indicating sensor failure. Tubing pressure fluctuated within a narrow range (2000–2100 psi), limiting the ability to create clean correlation of pressure and flow rates. Through multivariate analysis using AI, there was no linear correlation between production and FTP (Flowing Tubing Pressure). Nonetheless, pattern matching detected a few outlier days in which water cut growth was accompanied by pressure decline, worth exploring further. Such faint anomalies could not be detected with classical time-series plotting, suggesting the importance of machine-based pattern mining for noisy data. Gen AI models were used to simulate failure modes from oscillating flow behaviour.

These models predicted possible fluid interface movement in the wellbore and warned of early water breakthrough, which was later validated by trends in production logs. This also served to recognize that conventional diagnostics, particularly when utilizing erroneous sensor data, may miss early indications of production decline. The insight facilitated engineers in proposing improved shut-off designs and tubing pressure optimization to ensure stabilized production. This involved testing substitute flowing pressure

ranges and comparing simulated drawdown curves with operational history. The addition of Gen AI simulations provided the field team with several failure options and their relative probabilities under varying tubing configurations, enabling more confident operational choices. The knowledge enabled engineers to suggest enhanced shut-off plans and tubing pressure optimization to maintain stabilized production.

SE Texas Well

The second well, located in Southeast Texas, was also analyzed for gas-liquid separation inefficiencies and pressure-limited production ceilings. Approximately 60 days of daily production data, including oil, gas, water volumes, and wellhead pressures, were available. The well was known for high GOR and cyclic pressure variations, making it suitable for testing time-series models and generative backpressure simulations. AI modeling detected significant high GOR events that were not visible in typical plots. Clustering and time-series modeling revealed a periodic cycle of pressure swing that was occurring with oil production declines.

The cyclic behavior was a subtle one, frequently overlooked with rolling averages but distinctly detected through unsupervised learning models. Generative simulations recommended adjustment of the gas flow control valves to test controlled backpressure effect. These suggestions were validated by applying controlled choke operating adjustments and separator settings optimization. This resulted in stabilized production and improved oil cut within a 2-week span, primarily during peak operation. The AI dashboard facilitated near real-time visualization of shifting GOR trends and assisted field operators in monitoring intervention impacts on production KPIs. These changes lessened gas burden handling and enhanced tank-level balance. The faultless feedback loop between AI detection, Gen AI simulation, and field-level implementation was one step ahead in continuous production optimization.

NW India Well

The third example is from an aged onshore asset in Northwest India, with pump inefficiencies and high downtime being the norm. Approximately 90 days of pump speed, tubing pressure, and fluid level records were available. Frequent downtime and vibration-related shutdowns provided a challenging dataset where models needed to distinguish normal pump cycling from true efficiency losses. The AI pipeline took pump speed, fluid level, and tubing pressure inputs. AI-based anomaly detection identified inefficient pump cycles and trending pressure drawdown well before standard alarms triggered. The system also picked out pressure build-up intervals that showed early symptoms of constricted flow or equipment wear. Gen AI models experimented with different pump-off controller configurations in simulation to recommend optimum parameters for diverse loads. These were optimized set points for fluid level targets, pump runtime frequency, and controller feedback settings. Edge-deployed AI modules provided real-time alerts to operators when the pump was close to critical levels of vibration, avoiding mechanical failure. By doing this, pump efficiency improved and unscheduled interventions dropped by over 20% during the test duration. What made this case so useful was interpretability of AI forecasts operators weren't merely notified of impending failure risk, but also indicated the precise indicators (e.g., rising vibration frequency, declining drawdown gradient) responsible. This did much to enhance faith in the system, rendering adoption by field teams easier of revisions in operational practice.

Results and Discussions

The integration of AI and Gen AI workflows in the selected field cases yielded quantifiable gains in predictive maintenance, anomaly detection, and general production optimization. In all three wells, the AI-based models were able to discover latent inefficiencies and recommend operational changes much earlier than traditional monitoring equipment. These gains were not only quantitative but also process-related, affecting the manner in which field engineers understood and responded to real-time data.

In the KH#1 well of SW Louisiana, multivariate analysis using AI detected erratic responses from pressure gauges, particularly the SICP reading stuck at a uniform 125 psi on all but a few days. Although no such robust correlation was observed for Flowing Tubing Pressure (FTP) and production rate, AI models did pick up on underlying anomalies, namely those days when water production unexpectedly rose alongside muted pressure declines. These trends were not directly evident in conventional data visualization but were picked up by AI-based pattern discovery. As depicted in Figure 1, the multivariable analysis showed the absence of production variance in spite of changes in measured pressure values, whereas Figure 2 further established the absence of a direct proportionality between FTP and production output. Gen AI simulations supplemented this assessment by simulating flow instability and forecasting early indicators of water coning inside the wellbore. This information enabled the operations team to re-evaluate their tubing pressure approach and implement water shut-off methods more actively, enhancing stability and oil cut.

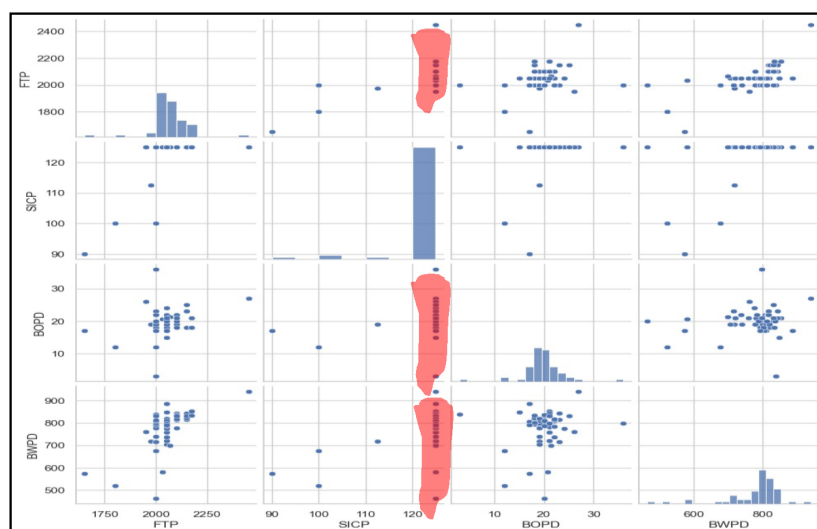


Figure 1—Multivariable plot showing pressure vs. oil and water production trends in KH#1 well.

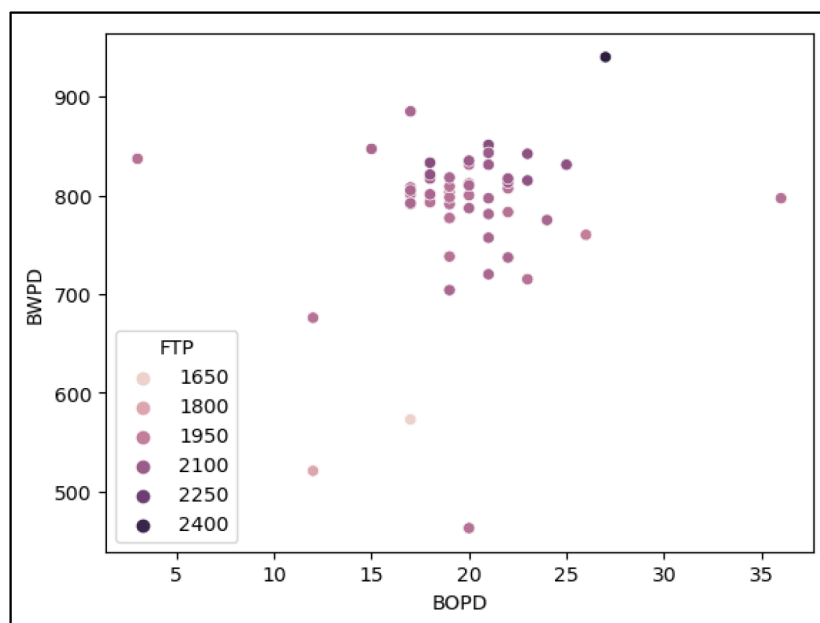


Figure 2—Relation between FTP and Production Rates

The challenges in the SE Texas well varied but were just as complicated. This strongly demonstrated high Gas-to-Oil Ratio (GOR) behavior, which was not explained by standard surveillance. The AI model detected recurring pressure oscillation cycles that closely followed decreasing oil production, revealing the relationship between erratic backpressure and suboptimal phase separation. Time-series modeling and clustering revealed these patterns, and Gen AI was employed to conduct simulated controlled backpressure scenarios to validate their impact on flow efficiency. The outcome of this simulation resulted in a field trial to which gas flow control valves were tweaked to apply backpressure regulation. In a two-week timeframe, oil production stabilized and GOR levels decreased significantly. Figure 3 shows the patterns of GOR spikes that became clearer with AI analysis. Field teams indicated enhanced control over separator tuning, while operators could respond to real-time alarms with improved context and confidence. The live, closed-loop feedback the cloud dashboard gave allowed these changes to be monitored constantly, and changing any aspect of operations became much simpler by observing the effects on production performance.

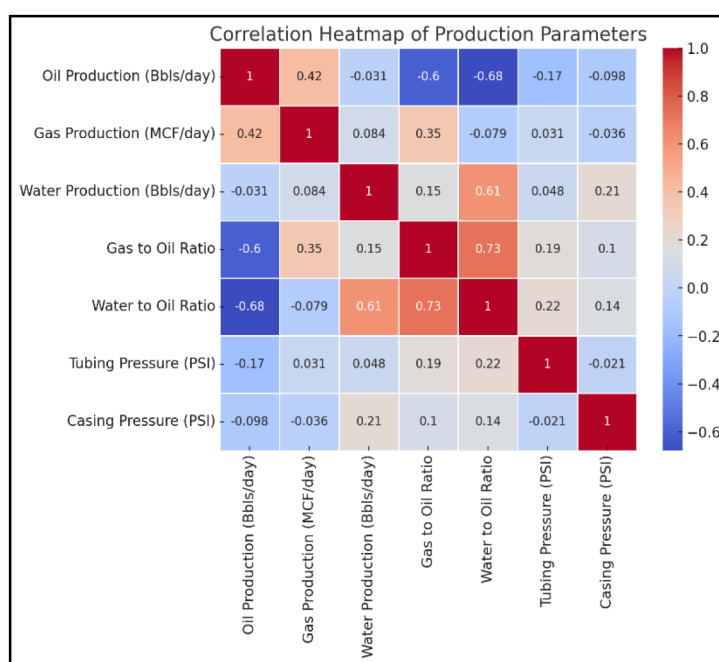


Figure 3—GOR variation pattern detected through AI-based analysis in SE Texas well.

In NW India well, situated in a mature, fully instrumented onshore field, the issue was a slightly different one of pump inefficiencies and downtime. AI models took in daily pump parameters like vibration frequency, tubing pressure, and fluid level, and could predict deteriorating pump performance long before conventional alarms were triggered. More significantly, the system provided interpretability showing which inputs like decreasing drawdown gradient or increased vibration levels were driving the deterioration trend. Gen AI models duplicated pump-off controller settings and runtime cycles across different operational loads to determine optimized configurations. Edge-based inference engines enabled this feedback to be sent to operators without latency delay. Figure 4 illustrates the correlation analysis of surface pressure readings with production output, confirming the correlation between tubing head pressure and oil rate. In accordance with AI suggestions, operators adjusted the pump schedule and target fluid levels, decreasing thereby manual interventions and enhancing considerably the reliability of pumps. During the deployment term, pump efficiency rose consistently, and downtime events decreased by over 20 percent.

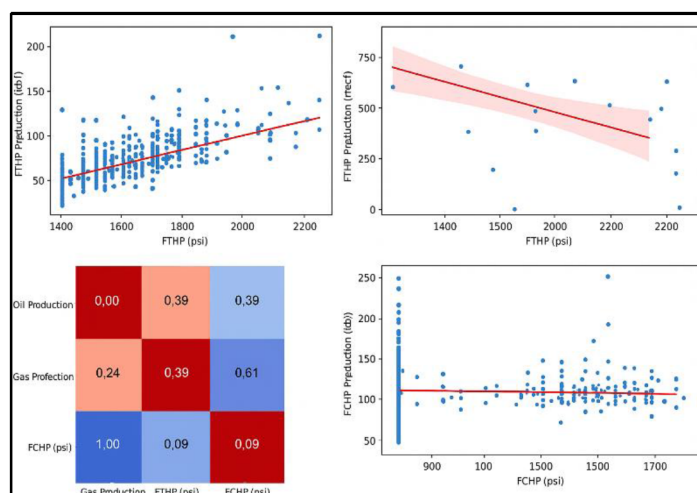


Figure 4—Correlation analysis of production variables from the NW India well. (a) FTHP vs. Oil Production showing a positive trend, (b) FTHP vs. Gas Production indicating a negative correlation, (c) Heat map revealing pairwise correlation among FTHP, FCHP, oil, and gas production, (d) FCHP vs. Oil Production showing weak association.

In all three examples, the combination of AI and Gen AI led to real-world operational gains. Predictive maintenance improved in accuracy, enabling more effective scheduling of field service. Downtime was minimized by identifying risk factors sooner, and production anomalies that would otherwise pass undetected were reported days ahead of time. The real-time dashboard also emerged as a valuable asset for the field engineers, who could engage with insights produced by AI instead of working only with static charts or historical norms. These outcomes not only proved the efficacy of marrying predictive analytics with generative simulation but also reflected the larger cultural change within operations from reactive to proactive decision-making. The explain ability of the system, providing reasoning behind each recommendation, also fostered adoption on both technical and non-technical teams. Through seamless integration into current workflows without interfering with field operations, the AI-Gen AI platform was shown to be scalable and feasible for actual oil and gas settings.

Conclusions

This research shows the potential impact of incorporating Artificial Intelligence (AI) and Generative AI (Gen AI) into oil and gas production operations. Using machine learning-based anomaly detection and scenario-driven simulation models, operators could predict failure, detect performance bottlenecks, and respond in advance to production issues. These findings were not restricted to theoretical verification but were realized in real-world field situations in three distinct wells from the United States and India, each with respective operating conditions and restrictions. The AI systems were able to identify potential problems that otherwise would have gone undetected through traditional monitoring like stuck sensor values, unoptimized pump cycles, or concealed gas-liquid handling inefficiencies adding real-time operational value. Most importantly, Gen AI usage provided an added layer to predictive maintenance by not just warning teams of future threats but also creating potential future conditions under different operating conditions.

The simulated scenarios allowed teams to virtually test "what-if" situations before implementing changes in the field, minimizing uncertainty and making more confident decisions. In each of the three wells, these combined strengths translated into quantifiable results: increased predictive accuracy, reduced unplanned shutdowns, and more consistent production profiles. Operators and field engineers started trending away from a reactive, event-driven work style and toward a more proactive, data-driven operational cadence. The effectiveness of strategy in this deployment was also linked to the way insights were presented.

With the use of edge-deployed inference engines and dashboards connected to the cloud, the system facilitated that information was received by the right individuals at the right time on the rig, in the control

room, or in a main office. Contrary to black-box AI systems, the models employed in this study yielded well-defined reasoning for each alert or suggestion. This interpretability was essential in gaining the trust of the operational teams and resulted in accelerated uptake of AI-aided workflows. In the future, this framework provides a scalable basis for larger digital transformation in oil and gas operations. As these technologies mature further, their coupling with real-time reservoir models, IoT sensor connections, and cloud-native asset management solutions can take their capabilities beyond production optimization. They can potentially enable safety monitoring, emissions monitoring, and even ESG reporting transforming conventional oilfields into smart, responsive assets. Finally, the deployment of AI and Gen AI in this manner is not merely a technological improvement, but rather a strategic shift toward more sustainable, efficient, and resilient oilfield operations.

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